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Source: *Journal of Marketing*, SEPTEMBER 2021, Vol. 85, No. 5 (SEPTEMBER 2021), pp. 74-94

Published by: Sage Publications, Inc. on behalf of American Marketing Association

Stable URL: https://www.jstor.org/stable/10.2307/27114322

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AMERICAN MARKETING ASSOCIATION

Journal of Marketing
2021, Vol. 85(5) 74–91
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DOI: 10.1177/0022242921997082
journals.sagepub.com/home/jmx
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When Algorithms Fail: Consumers' Responses to Brand Harm Crises Caused by Algorithm Errors

Raji Srinivasan and Gülen Sarial-Abi

Abstract

Algorithms, increasingly used by brands, sometimes fail to perform as expected or, even worse, cause harm, leading to brand harm crises. Algorithm failures are unfortunately increasing in frequency, yet little is known about consumers' responses to brands following such crises. Extending developments in the theory of mind perception, the authors hypothesize that, following a brand harm crisis, consumers respond less negatively if the error was caused by an algorithm (vs. a human). The authors further hypothesize that consumers' lower mind perception of agency of the algorithm (vs. a human), which lowers their perceptions of the algorithm's responsibility for the harm, mediates this relationship. Four moderators of this relationship are hypothesized: two algorithm characteristics (whether the algorithm is anthropomorphized and whether it involves machine learning) and two characteristics of the task for which the algorithm is deployed (whether the task is subjective [vs. objective] and whether it is interactive [vs. noninteractive]). The authors find support for the hypotheses in eight experimental studies. The effects of two managerial interventions to manage brand harm crises caused by algorithm errors are examined. This research advances the literature on brand harm crises, algorithm usage, and algorithmic marketing and generates managerial guidelines to address such crises.

Keywords

algorithm errors, algorithmic marketing, brand harm crises, theory of mind perception

Online supplement: https://doi.org/10.1177/0022242921997082

Given the explosive growth in the volume of data, dramatic developments in software programs, and the decreasing cost of cloud computing, the use of algorithms—software programs that organize data, predictions, and decisions—has grown exponentially. Although this has occurred across many contexts, algorithm usage in the marketing context—algorithmic marketing—has increased dramatically. Algorithmic marketing has many advantages, including lower costs, high efficiency, and high effectiveness (Gal and Elkin-Koren 2017). Despite these advantages, though, there is growing evidence of algorithm failures across multiple contexts (Griffith 2017). In marketing, algorithm errors harm consumers and/or violate consumers' expectations of the brand's values, creating brand harm crises. In a survey of chief marketing officers (CMOs), fielded by the CMO Council and Dow Jones Inc., most CMOs (78%) expressed concern about the threats to their brands' reputations from algorithm errors (Vizard 2017).

Although algorithms operate in the digital domain, algorithm errors have many real-world consequences, including causing substantive harm to brands. We discuss two examples to provide additional context. First, there is evidence (Diakopoulos 2013; Sweeney 2013) of algorithmic defamation in online searches. Algorithm-based Google search autocompletion routines make incorrect defamatory associations about groups of people (Badger 2019). For example, searching for certain ethnic names on Google provides results including advertisements for bail bonds or criminal record checking. Second, Apple Credit Card, launched in partnership by Apple Inc. and Goldman Sachs Inc. in August 2019, faced reputational harm when users noticed that it offered lower lines of credit to women than to men of equal or even lower financial standing (Vigdor 2019). In response, the New York Department of Financial Services announced an investigation of Apple Inc. to assess a breach of federal financial rules regarding equal financial access. Cognizant of the potential harm that could be caused by algorithm errors, for the first time, Google's

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parent company, Alphabet Inc. (February 2019), and Microsoft Inc. (August 2018) acknowledged in their annual reports that "flawed" algorithms could result in "brand or reputational harm" and have an "adverse effect" on financial performance (Vincent 2019). In summary, algorithm errors are a key and growing source of brand harm crises.

Brand harm crises are adverse negative events inconsistent with a brand's values. In a brand harm crisis, the brand's ability to deliver promised benefits to consumers is compromised or, even worse, the brand causes physical harm to consumers (Dutta and Pullig 2011; Pullig, Netemeyer, and Biswas 2006), causing consumers to respond negatively to the brand (Ahluwalia, Burnkrant, and Unnava 2000; Lei, Dawar, and Gürhan-Canli 2012; Swaminathan, Page, and Gürhan-Canli 2007). Consumers' attributions about what caused the harm influence their subsequent responses to the brand (Folkes 1984; 1990). Consumers feel angry and seek revenge if they believe that the firm was responsible for the harm and could have prevented it (Folkes, Koletsky, and Graham 1987). See Cleeren, Dekimpe, and Heerde (2017) for a comprehensive review of the brand harm crises literature. Because the growth in algorithmic marketing is fairly recent, research has not yet addressed harm crises caused by algorithm errors.

There is a large body of research in multiple literatures, including in marketing, on people's responses to nonhuman agents (e.g., algorithms, computers, robots). People treat computers as social actors although they know that computers do not possess feelings, intentions, motivations, or "selves" (Moon 2000; Nass and Moon 2000). Other work (Choi, Matilla, and Bolton 2020) suggests that humanoid (vs. nonhumanoid) service robots are more strongly associated with warmth (but not competence).

Past work on algorithm usage has examined people's responses to using algorithms (Castelo, Bos, and Lehmann 2019; Dietvorst, Simmons, and Massey 2015). Individuals prefer doing a task themselves or having it done by their peers with whom they have more in common (Prahl and Van Swol 2017), rather than using an imperfect algorithm (i.e., people display algorithm aversion [Dietvorst, Simmons, and Massey 2015]). This preference for humans over algorithms persists even when it worsens outcomes. In contrast, in the advice-giving context (absent of algorithm errors), Logg, Minson, and Moore (2019) report algorithm appreciation (i.e., people incorporate advice from algorithms more than from humans). Related recent work on automated vehicles operated by algorithms (Awad et al. 2020; Gill 2020) suggests that individuals consider harm to pedestrians by an automated vehicle (vs. themselves as the driver in a regular car) to be more permissible. See Castelo, Bos, and Lehmann (2019) for a good overview of the research on algorithm usage (specifically Table 1).

In summary, past research on algorithms has overlooked how consumers respond to a brand following a brand harm crisis caused by an algorithm error (vs. human error), which is the focus of this research. Distinct from past research on algorithm usage that considers the individual's decision to use the algorithm, in this research context, the decision to use the algorithm is taken by the brand manager rather than by the consumer who experiences the harm caused by the algorithm error. Furthermore, the dependent variable here is the consumer's response to the brand and not to the algorithm that commits the error, which has been the focus of past research on algorithm usage. Furthermore, we examine the moderation effects of two algorithm characteristics and two characteristics of the task in which the error occurs on this relationship. As consumers' responses to a brand harm crisis are always negative (Lei, Dawar, and Gürhan-Canli 2012), we examine consumers' negative responses to a brand following a brand harm crisis caused by an algorithm error.

We apply the theory of mind perception (Epley and Waytz 2009; Gray, Gray, and Wegner 2007; Gray and Wegner 2012), which proposes that individuals ascribe minds to other entities (e.g., individuals, animals, robots) and consider the contents of these entities' minds. Specifically, we examine consumers' mind perception of the algorithm's agency (i.e., the error-committing entity's perceived capacity to intend and to act).

Features of an entity can change people's mind perception of the entity's agency (Waytz, Cacioppo, and Epley 2010). Accordingly, we hypothesize that, following a brand harm crisis caused by an algorithm error (vs. human error), consumers will, ceteris paribus, have lower mind perception of agency of the algorithm (the entity) and assign it lower responsibility for the harm caused, weakening their negative responses to the brand. Furthermore, individuals' responses to an algorithm vary depending on task characteristics (Castello et al. 2019). Accordingly, we consider four moderators of consumers' responses to a brand following a harm crisis caused by an algorithm error: two algorithm characteristics (whether the algorithm is anthropomorphized and whether it involves machine learning) and two characteristics of the task in which the algorithm error occurs (whether the task is subjective [vs. objective] and whether it is interactive [vs. noninteractive]). We test and find support for the hypotheses in eight experimental studies, including an incentive-compatible study with a consequential outcome (donation to a charity) and two studies with behavioral measures.

This research's insights extend the literature on harm crises by studying an inanimate source of errors, algorithms, which have hitherto been overlooked as a source of brand harm crises in the marketing literature. Second, in a novel extension to the algorithm usage literature, which has thus far focused on consumers' responses to an algorithm, we find that consumers' responses to the brand are more forgiving of algorithm errors when they do not have the authority to choose whether to use the algorithm. Third, we identify consumers' mind perception of algorithms' agency as a potential key building block, relevant in the development of a theory of algorithmic marketing. Fourth, by identifying the moderating role of algorithm and task characteristics, this research's insights make a novel contribution to the harm crises literature, which has not examined characteristics of the sources of the error and the task as factors affecting outcomes in harm crises. Using insights from the findings of the studies involving the four moderators and a

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Table 1. Overview of Studies.

Study	Participants	Context; Dependent Variable (DV)	Conditions and Results				Conclusion
Prestudy	N = 403, online	Financial investments; DV: brand attitude	Error		No error		When there is no error, there is no difference in consumers' responses to a brand that uses an algorithm versus a human.
			Algorithm 4.55 (1.56)[5]	Human 3.63 (1.79)	Algorithm 5.55 (1.03)	Human 5.31 (1.07)	As hypothesized in H_1, consumers' responses to a brand following a brand harm crisis are less negative when the error is caused by an algorithm (vs. human).
1a	N = 157, online	Mistake in headline of a fundraising advertisement; DV: amount of donation	Algorithm error 20.71 (41.91)		Human error 7.84 (22.34)		Support for H_1.
1b	N = 233, online	Online platform made a mistake and provided bad financial advice; DV: advice provided	Algorithm error 3.19 (2.18)		Human error 2.49 (1.91)		Support for H_1.
1c	N = 177, U.S. undergraduate students	Glitch in the Qualtrics online computer system; DV: % intention to re-engage with the brand	Algorithm error 64.3%		Human error 42.1%		Support for H_1.
2	N = 251, online	Recall of 4.8 million vehicles; DV: brand attitude mediators: (1) mind perception of source of the error's agency in committing the error and (2) perceptions of source of the error's responsibility for the harm caused to the brand.	Algorithm error 4.59 (1.61)		Human error 4.17 (1.69)		Support for H_1, H_2, and for serial mediation by lower agency of the algorithm and responsibility for the harm caused by the algorithm error.
3	N = 372, online	Mistake in investment decisions for customers of a financial investment company; DVs: brand attitude, donation amount	Human error 3.63 (1.81) 156.88 (165.19)		Algorithm error 4.25 (1.84) 204.98 (180.05)	Anthropomorphized algorithm Error 3.74 (1.76) 160.40 (157.47)	As hypothesized in H_3, consumers' responses to a brand following a brand harm crisis caused by an algorithm error are more negative when the error is caused by an anthropomorphized (vs. not) algorithm.
4	N = 310, online	Mistake in users' Twitter timelines; DV: brand attitude	Human error 4.20 (1.47)		Algorithm error 4.76 (1.62)	Machine learning algorithm error 4.21 (1.45)	As hypothesized in H_4, consumers' responses to a brand following a brand harm crisis caused by an algorithm error are more negative when the error is caused by a machine learning (vs. not) algorithm.
5	N = 400, online	A leading university in the United States faced a crisis because of an error in the subjective (vs. objective) assessment of Asian American students' applications DV: brand attitude	Algorithm error		Human error		As hypothesized in H_5, consumers' responses to a brand following a brand harm crisis caused by an algorithm error are more negative when the algorithm error occurs during a subjective (vs. objective) task.
			Subjective task 3.76 (1.64)	Objective task 4.46(1.49)	Subjective task 4.28 (1.73)	Objective task 3.99 (1.64)	
6	N = 328, U.S. undergraduate students	Mistake in product selection by a personal stylist of a fashion retailer company DV: brand attitude	Algorithm error		Human error		As hypothesized in H_6, consumers' responses to a brand following a brand harm crisis caused by an algorithm error are more negative when the error occurs during an interactive (vs. noninteractive) task.
			Interactive task 3.41 (1.05)	Noninteractive task 3.82(1.22)	Interactive task 3.48 (.98)	Noninteractive task 3.37 (.95)	
M1	N = 368, online	Recall of 4.8 million vehicles DV: brand attitude	Algorithm error		Human error		Responses to a brand following a brand harm crisis caused by an algorithm error are more negative when there is more human supervision (vs. technological supervision).
			Technological supervision 4.71 (1.69)	Human supervision 4.13 (1.56)	Technological supervision 4.15 (1.65)	Human supervision 4.63 (1.69)	

Note: Figures in parentheses are standard deviations.

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managerial interventions study, we provide guidance to managers on the deployment of algorithms given their effects on consumers' responses when they commit errors, as well as how to manage the aftermath of such brand harm crises.

Theory

Early work on people's responses to nonhuman agents (e.g., computers) suggests that consumers mindlessly apply social norms (Moon 2000) in their interactions with such agents, including displaying a self-serving bias in attributions of responsibility to positive versus negative service encounters (Moon 2003). Building on these ideas, we apply the theory of mind perception in the psychology literature (Gray, Gray, and Wegner 2007; Gray and Wegner 2012) about people's perceptions of the minds of other entities to algorithms to develop the hypotheses. We first provide a brief overview of the theory of mind perception and then develop the hypotheses.

Theory of Mind Perception: A Brief Overview

Mind perception, also known as humanizing or mentalizing, involves individuals making inferences about their own and others' (entities) mental states by positing unobservable properties such as intentions, desires, goals, beliefs, and secondary emotions to serve as mediators between their sensory inputs and subsequent actions (Gray, Gray, and Wegner 2007; Gray and Wegner 2012). According to the theory of mind perception, a perceiver needs to implicitly determine the extent to which an entity has a mind and then determine that entity's state of mind. In addition to perceiving the minds of other humans, people are capable of perceiving minds in nonhuman entities such as animals, gadgets, or software.

People perceive other entities' minds in terms of two psychological capacities: agency and experience (Gray, Gray, and Wegner 2007). Mind perception of the entity's agency is its perceived capacity to intend and to act (e.g., self-control, judgment, communication, thought, memory), and mind perception of experience is the entity's perceived capacity for sensation and feeling (e.g., hunger, fear, pain, pleasure, consciousness). When discussing mind perception of agency, Gray, Gray, and Wegner (2007) posit that agency qualifies entities as moral agents, capable of reasoned actions and having the capacity to do right or wrong (Gray and Wegner 2009; Gray, Young, and Waytz 2012), whereas experience qualifies entities as moral patients, capable of benefiting from good or suffering from evil acted upon them.[1]

In this research, we consider consumers' mind perception of agency of the algorithm that has committed the error and do not consider consumers' mind perception of the algorithm's experience as a moral patient being acted upon by others, which is not relevant when the algorithm commits errors. We note that individuals' mind perception of agency of an entity are positively related to judgments of the entity's responsibility for harm caused (Waytz, Heafner, and Epley 2014), which is consistent with common law practice that holds individuals with diminished mental capacity as being less responsible for their transgressions.

Overview of Hypotheses

We propose that, following a brand harm crisis caused by an algorithm error (vs. human error), consumers will have lower mind perception of agency of the algorithm (than they would for humans) for the error and assign lower responsibility to the algorithm for the harm caused, resulting in a less negative response to the brand.

Features of the entity can change people's mind perception of its agency (Waytz, Cacioppo, and Epley 2010). Furthermore, individuals' responses to algorithms vary depending on the characteristics of the task for which the algorithm is deployed (Logg, Minson, and Moore 2019). Extending these two ideas, we propose four factors that will moderate consumers' responses to a brand following a brand harm crisis caused by an algorithm error: two algorithm characteristics (whether the algorithm is anthropomorphized and whether it involves machine learning) and two characteristics of the task in which the error occurs (whether the task is subjective [vs. objective] and whether it is interactive [vs. noninteractive]).

Main effect of algorithm (vs. human) error. An entity's mind perception of agency to intend and to act affects individuals' perception of the entity's responsibility for its actions. For example, people have lower mind perception of agency of an inanimate robot than of a man or of a young girl (Gray, Gray, and Wegner 2007), suggesting that they would perceive lower responsibility for a robot's harmful actions. Extending this idea to algorithm errors, we propose that people will have lower mind perception of the agency of the algorithm (vs. a human) that commits the error responsible for the brand harm crisis and assign lower responsibility[2] to the algorithm for the harm caused.

The notion that people consider algorithms to have lower agency than the humans who developed them is consistent with early research on individuals' interactions with computers and robots (Moon 2000) and the recent research on algorithm aversion (Dietvorst, Simmons, and Massey 2015) and algorithm appreciation (Logg, Minson, and Moore 2019; Prahl and Van

[1] The common everyday meaning of "experience" as "practical contact with and observation of facts or events" (from Merriam-Webster) is distinct from the use of the term "experience" in the theory of mind perceptions, defined as the capacity for sensations (i.e., algorithms being capable of feeling).

[2] We note that the meaning of the term "responsibility" has three commonplace meanings (from Merriam-Webster): (1) the state or fact of having a duty to deal with something or of having control over someone, (2) the state or fact of being accountable or to blame for something, and (3) the opportunity or ability to act independently and make decisions without authorization. Our use of the term "responsibility" is as per the second definition. Consistent with this interpretation, "blame" is a synonym for "responsibility" at https://www.merriam-webster.com/dictionary/responsibility. Thus, our view of responsibility for the harm is consistent with blame for the harm.

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Swol 2017). This argument is consistent with other evidence about algorithms (McCullom 2017) showing that because algorithms do not have "human-like" qualities, people may not hold them fully responsible for actions that cause harm.

Accordingly, we propose that consumers' responses to a brand following a brand harm crisis caused by an algorithm error (vs. human error) will be less negative. We further propose that consumers' responses to the brand will be serially mediated by their lower mind perception of the algorithm's agency, which, in turn, will lower their perceptions of the algorithm's responsibility for the harm caused by the error. Thus, we propose:

> **H_1:** Consumers' responses to a brand following a brand harm crisis caused by an algorithm error (vs. human error) are less negative.
>
> **H_2:** Consumers' lower mind perception of the algorithm's agency, which lowers their perceptions of the algorithm's responsibility for the harm caused by the error, mediates the relationship in H_1.

Anthropomorphized algorithm. Anthropomorphism is the process of inductive inference in which people attribute distinctively human characteristics to inanimate objects, including brands, machines, technologies, and software (Kim and McGill 2011). Anthropomorphizing an entity involves the use of human traits (e.g., humanlike face and name) so that individuals attribute essential human characteristics (e.g., humanlike mind capable of thinking and feeling) to the entity. A common marketing practice is to name products with human names with the intent of anthropomorphization (e.g., IBM's artificial intelligence software "Watson," Bank of America's virtual financial assistant "Erica," Amazon's virtual assistant "Alexa").

The effects of anthropomorphization on consumer behaviors have received attention from marketing scholars (Aggarwal and McGill 2007; 2012; Kim and Kramer 2015). The overall evidence suggests that the higher a product's anthropomorphization, the higher consumers' evaluations of it (Aggarwal and McGill 2007) and the higher its sales (Landwehr, McGill, and Herrmann 2011). With regard to harm crises, anthropomorphization of a product lowers consumers' evaluations of it (Puzakova, Kwak, and Rocereto 2013), which is consistent with the main effect in H_1.

In the technology context, relevant to this research, firms anthropomorphize products to make them user-friendly and less intimidating (Lafrance 2014). Anthropomorphizing technology-driven products increases consumers' positive feelings toward the products, reduces people's fear of technology, and suggests that the products can perform their intended functions well (Waytz, Heafner, and Epley 2014). This results in consumers assigning higher responsibility to anthropomorphized products, indeed, at a level comparable to what they assign to humans (Epley, Caruso, and Bazerman 2006).

Accordingly, we suggest that when an anthropomorphized (vs. not) algorithm is the source of the error that causes a brand harm crisis, consumers will consider the anthropomorphized algorithm to have higher mind perception of agency and assign higher responsibility to it for the harm caused by the algorithm error. We hypothesize that consumers' responses to a brand following a brand harm crisis caused by an algorithm error will be more negative when the algorithm is anthropomorphized (vs. not). Thus, we propose:

> **H_3:** Consumers' responses to a brand following a brand harm crisis are more negative when the error is caused by an anthropomorphized (vs. not) algorithm.

Machine learning algorithm. Machine learning algorithms learn "by themselves" (i.e., independently using historical data, models, and analyses). In other words, machine learning algorithms are programmed such that they can modify themselves (i.e., without human intervention) to improve their performance. The availability of "big data," growing computational power, and developments in software technology enable such machine learning algorithms to learn independently from their experiences working repeatedly on large data sets (Heller 2019). Machine learning algorithms know users' behaviors and leverage that knowledge to recommend products that match users' preferences. Such machine learning algorithms power Amazon, Netflix, and Spotify recommendations; Google Maps; and much of the content on Facebook, Instagram, and Twitter.

Developments in bioethics consider an entity's capacity for learning, including the ability to think, reason, and remember, to be indicative of superior mental abilities and to define the degree of its humanness (Fletcher 1979). Reiterating this view, Gray and Wegner (2009) compared people's perceptions of mentally competent (vs. mentally challenged) adults and found them to be higher on mental abilities associated with learning and mind perception of agency.

Applying these ideas, we propose that consumers will ascribe more humanness to a machine learning (vs. not) algorithm. Following a brand harm crisis resulting from an error caused by a machine learning (vs. not) algorithm, people may perceive the machine learning algorithm to have higher agency and, therefore, higher responsibility for the harm caused. Thus, we hypothesize that, following a brand harm crisis caused by an error of a machine learning (vs. not) algorithm, consumers' responses to the brand will be more negative. Thus, we propose:

> **H_4:** Consumers' responses to a brand following a brand harm crisis are more negative when the error is caused by a machine learning (vs. not) algorithm.

Subjective (vs. objective) task. Following Castelo, Bos, and Lehmann (2019), a subjective task is open to interpretation on the basis of an individual's personal opinion, whereas an objective task is one that involves factors that are quantifiable and measurable. People perceive subjective tasks as requiring intuition and objective tasks as requiring human traits such as logical, rule-based analysis (Inbar, Cone, and Gilovich 2010). Although algorithms are proficient at objective tasks, the

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growth of big data and lower costs of computing have resulted in a dramatic increase in the use of algorithms for subjective tasks (Kleinberg et al. 2018). Companies routinely use algorithms for subjective tasks, such as selecting applicants (e.g., Indeed.com, university admissions) and recommending clothing for consumers (e.g., J. Jills, Stitchfix.com). Ceteris paribus, consumers perceive that algorithms lack the ability to perform subjective tasks (Castelo, Bos, and Lehmann 2019), although making an algorithm more humanlike is effective at increasing its usage for subjective tasks. In other words, individuals ascribe higher humanness to an algorithm deployed for subjective tasks.

Applying this logic, we propose that when an algorithm is used for a subjective (vs. objective) task that requires intuition and an algorithm error causes the brand harm crisis, consumers will perceive the algorithm as having higher mind perception of agency and hold it more responsible for the harm caused. Therefore, we posit that in a brand harm crisis caused by an algorithm error, when the algorithm error occurs during a subjective (vs. objective) task, consumers' responses to the brand will be more negative. Thus, we propose:

> **H_5:** Consumers' responses to a brand following a brand harm crisis are more negative when the algorithm error occurs during a subjective (vs. objective) task.

Interactive (vs. noninteractive) task. A key characteristic of interactive communications between entities (say, a human and a computer) is contingency in responses (Sundar 2009). In an interactive communication (Sundar et al. 2016), each entity acknowledges and incorporates the other entity's prior communications. Higher interactivity between two individuals in an online context heightens perceptions of each other's humanness (Sundar et al. 2015). Interactivity between an individual and a nonhuman entity (e.g., an algorithm) makes the entity more humanlike because it mimics the contingency in real-time interactive exchanges between humans (Rafaeli 1988). Therefore, people may perceive the algorithm in an interactive task as being capable of communication, an integral aspect of people's mind perception of agency of an entity (Gray, Young, and Waytz 2012). Indeed, algorithms are now widely used by marketers in interactive communications including in customer service chatbots (e.g., Spotify) and product recommendations (e.g., Stitchfix).

Applying these ideas, we anticipate that consumers will have higher mind perception of agency when the task in which the algorithm error occurs is interactive (vs. noninteractive) between consumers and the algorithm. We propose that, following a brand harm crisis caused by an algorithm error in an interactive (vs. noninteractive) task, consumers will hold the algorithm more responsible for the harm, causing their responses to the brand to be more negative. Thus, we propose:

> **H_6:** Consumers' responses to a brand following a brand harm crisis caused by an algorithm error are more negative when the error occurs in an interactive (vs. noninteractive) task.

Prestudy

We conducted a prestudy that examined consumers' responses to a brand when there are no errors to ensure that the effects we theorize relate only to the error caused by the algorithm (vs. human) and not to algorithms in general.[3] We preregistered this prestudy on AsPredicted.org (#43436). We provide stimuli for the prestudy and all other studies in the Web Appendix and summary of the studies and findings in Table 1.

Participants and Procedure

A total of 403 adults participated in the experiment on Amazon Mechanical Turk (MTurk) in exchange for $.50 (219 male; $M_{age} = 37.73$ years, SD = 12.40). The study used an error (vs. no error) and algorithm (vs. human) between-subjects design.

We randomly assigned participants to error and "no error" conditions. Participants in the error condition read that HMS Investments, a leading financial investment company, was facing a crisis. In the "no error" condition, participants read that HMS Investments reduces risks for its clients. We also randomly assigned participants to algorithm and human conditions. Participants in the algorithm (human) error condition read that HMS Investments, a leading financial investment company, was facing a crisis because a financial algorithm program (financial manager) had committed an error, resulting in financial losses for its customers. Participants in the algorithm (human) "no error" condition read that HMS Investments reduces risks for its clients with its strong computer algorithms (employees). We measured participants' attitude toward HMS Investments using a five-item scale (Ahluwalia, Burnkrant, and Unnava 2000; Swaminathan, Page, and Gürhan-Canli 2007): "bad/good," "low quality/high quality," "undesirable/desirable," "harmful/beneficial," and "unfavorable/favorable" ($\alpha = .96$). Participants then provided their basic demographic information.

[3] The Institutional Review Board of the authors' home institutions reviewed and approved the experimental design before commencing the research. Participants in all studies provided informed consent before participation.As an empirical practice, we had a rule of thumb of ensuring at least 30 participants per cell for lab studies and at least 75 participants per cell for online studies. For the lab studies using student participants, we did not know, a priori, the number of participants. We report all variables collected and all conditions in the studies and do not exclude data from the analyses unless otherwise noted for clearly identified reasons. We report the number of excluded participants and do not add data from additional participants in any study, following the analyses.We preregistered analyses (and exclusions) at AsPredicted.org for the prestudy as well as Studies 3, 5, and the managerial intervention study. We conducted Studies 1a–1c, 2, 4, and 6 before preregistration became our standard practice. Anonymized links to the preregistrations of studies are available on request.We conducted all analyses on SPSS Statistics 23 and 25 IBM software.

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Results

Brand attitude. An analysis of variance (ANOVA) on brand attitude reveals the predicted interaction effect of error (vs. no error) and algorithm (vs. human) conditions, $F(1, 399) = 5.86$, $p = .016$. There is no main effect of error (vs. no error), $p = .157$, and algorithm (vs. human) conditions, $p = .335$. Participants' responses to a brand following a brand harm crisis caused by an algorithm (vs. human) error are less negative: $M_{AE} = 4.55$, $SD = 1.56$ versus $M_{HE} = 3.63$, $SD = 1.79$, $F(1, 399) = 21.63$, $p < .001$. However, when there is no error, there is no difference in participants' responses when the brand uses algorithms versus when they use humans: $M_{algorithm} = 5.55$, $SD = 1.03$ versus $M_{human} = 5.31$, $SD = 1.07$, $F(1, 399) = 1.53$, $p = .217$. There is no effect of age, $p = .085$, or gender, $p = .612$.

Thus, when there is no error, consumers' responses to a brand are similar regardless of whether the brand uses an algorithm or a human. However, as hypothesized in H_1, consumers' responses to a brand following a brand harm crisis caused by an algorithm (vs. human) error are less negative. These findings provide preliminary support for H_1 and indicate that algorithm error, and not the mere presence of the algorithm, drives the results in the subsequent studies.

Studies 1a–1c: Main Effect of Algorithm Error (vs. Human Error)

Study 1a

In Study 1a, we examine consumers' responses to a brand following a brand harm crisis caused by an algorithm error (vs. human error) with an incentive-compatible experiment using a consequential outcome (donation to a charity suggested by the brand) as the dependent variable. A lower donation to the charity denotes a more negative response to the brand.

Participants read about a consumer electronics retailer for which an algorithm error or a human error had caused a brand harm crisis. Participants then indicated the amount that they were willing to donate to the World Health Organization through the electronics retailer from the compensation that they would receive in the study.

Participants and procedure. A total of 157 U.S. adults participated in the experiment on MTurk in exchange for $1.50 (84 male; $M_{age} = 40.69$ years, $SD = 10.37$). All participants read that a consumer electronics company, Qualtronics, was facing a harm crisis. This was because their fundraising campaign, BanishCovid19, aimed at combating the COVID-19 pandemic, implied that a Chinese virus caused COVID-19. The fundraising campaign was for the World Health Organization.

Participants in the algorithm error condition read, "because the disease was first detected in Wuhan Province of China, Qualtronics used computer algorithms to design the advertisement and released the campaign with the headline 'Contribute to BanishCovid19 and Destroy the Chinese Virus.'" Participants in the human error condition read, "because the disease was first detected in Wuhan Province of China, Qualtronics' managers designed the advertisement and released the campaign with the headline 'Contribute to BanishCovid19 and Destroy the Chinese Virus.'" In both conditions, participants read that following negative feedback from their customers, Qualtronics apologized to its customers and changed the advertisement headline to "Contribute to BanishCovid19 and Destroy the Corona Virus."

We then provided participants in both conditions the opportunity to donate to the World Health Organization through Qualtronics. The maximum amount they could donate was the $1.50 they would earn in the study. Participants indicated the amount they would donate on a sliding scale ($M = 14.40$ cents, $SD = 34.47$ cents).

As a manipulation check, participants indicated the extent to which they believed the error was caused by a human in Qualtronics (1 = "not at all," and 7 = "very much"). Participants also indicated the extent to which they were concerned about COVID-19 and the extent to which COVID-19 had impacted their community on seven-point scales (1 = "not at all," and 7 = "very much"). Participants then provided their basic demographic information, including race.

Results

Manipulation check. As intended, participants in the human error (vs. algorithm error) condition indicated that the source of the error in Qualtronics is more human: $M_{HE} = 6.29$, $SD = 1.34$ versus $M_{AE} = 5.81$, $SD = 1.57$, $t(155) = 2.03$, $p = .044$.

Amount of donation. The results indicate a significant effect of algorithm error (vs. human error) condition on the donation amount: $M_{AE} = 20.71$ cents, $SD = 41.91$ cents versus $M_{HE} = 7.84$ cents, $SD = 22.34$ cents, $F(1, 155) = 5.70$, $p = .018$. When we included the three control variables of participants' race, concerns about COVID-19, and COVID-19's impact on their community in the model, the effect of algorithm error (vs. human error) on the amount of the donation is still significant, $F(1, 152) = 5.11$, $p = .024$. There is no main effect of the three control variables or of age, $p = .289$, or gender, $p = .202$.

In Study 1a, consumers' donations following a brand harm crisis caused by an algorithm error (vs. human error) are higher. This finding supports the prediction (H_1) that consumers' responses to the brand following a brand harm crisis caused by an algorithm error (vs. human error) are less negative.

Study 1b

In Study 1b, we examine consumers' behavioral responses to a brand harm crisis caused by an algorithm error (vs. human error) at a fictitious global platform company, Life Skills Without Borders, an online advice crowdsourcing website for young adults. We randomly assigned participants to either the algorithm error or human error condition and measured the number of items of advice provided by participants to Life Skills Without Borders following a brand harm crisis.

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Participants and procedure. The experiment used the algorithm error (vs. human error) condition as a between-subjects design. A total of 233 participants participated in the experiment on the Prolific online platform in exchange for one British pound (101 male; $M_{age} = 35.89$ years, SD = 12.36).

All participants read about Life Skills Without Borders, a global crowdsourcing platform for providing life skills advice to young adults. We randomly assigned participants to either the algorithm error or human error conditions. Participants in the algorithm error (human error) condition read that a computer algorithm (an employee) at Life Skills Without Borders had made a mistake and provided bad financial advice to poor young couples, resulting in financial losses.

We then informed participants that Life Skills Without Borders was presently crowdsourcing ideas for providing post-graduation career advice to young adults. We asked participants to provide advice to Life Skills Without Borders, which was the study's dependent variable. We used the number of unique items of advice provided by each participant (e.g., "follow your heart and work at a job that you think you might like," "if you get an offer for a higher paying job at a different company be sure you want the job before you take it") as the dependent variable. Because the only difference between the two (between-subjects) conditions was the source of the error (algorithm vs. human), we considered higher numbers of pieces of advice provided by participants as indicative of participants' less negative response to Life Skills Without Borders. Participants then provided their basic demographic information.

Results and discussion. One of the authors coded the number of unique items of advice provided by the participants ($M_{advice\#} = 2.52$, SD = 2.13). A t-test shows that participants in the algorithm error condition provided more advice than participants in the human error condition: $M_{AE} = 3.19$, SD = 2.18 versus $M_{HE} = 2.49$, SD = 1.91, $t(229) = 72.56$, $p = .011$. The results of Study 1b support our prediction (H_1) that consumers' behavioral responses to a brand following a brand harm crisis caused by an algorithm error (vs. human error) are less negative.

Study 1c

In Study 1c (details in the Web Appendix), we examine consumers' reengagement behaviors with the brand following a brand harm crisis caused by a failure in an online computer system. An algorithm error (vs. human error) disrupted an online task in Qualtrics, a software program used for lab experiments. We randomly assigned participants either to the algorithm error or human error condition and noted their decision to repeat or not repeat the online task (i.e., reengage with Qualtrics). Participants' willingness to repeat the task would indicate a less negative response to the Qualtrics brand. The results support our prediction (H_1) that consumers' reengagement behaviors with the brand (i.e., whether they would repeat the online task) following a brand harm crisis are less negative when caused by an algorithm error (vs. human error).

Study 2: Mediation by Mind Perception of Algorithm's Agency and Responsibility for Harm

In Study 2, we examine the role of consumers' mind perception of the source of the error's agency in committing the error and responsibility for the harm caused in serially mediating consumers' responses to the brand following a brand harm crisis caused by an algorithm error (vs. human error) (H_2).[4] For a test of the mediation (H_2), we measured participants' mind perception of agency of the source of the error that caused the brand harm crisis and perceptions of the source of the error's responsibility for the harm caused by the error.

Participants and Procedure

A total of 251 adults participated in the between-subjects experiment on the MTurk online platform in exchange for $.50 (137 male; $M_{age} = 34.98$ years, SD = 11.19). All participants saw a tweet on the official Twitter account of *The New York Times* announcing the recall of 4.8 million Fiat Chrysler vehicles because of a cruise control problem. We randomly assigned participants to either the algorithm error or human error condition. Participants in the algorithm error (human error) condition read that a Fiat Chrysler computer algorithm (Fiat Chrysler employees) had made a mistake resulting in a defect in the cruise control system, causing a safety hazard.

We measured participants' attitude toward the Fiat Chrysler brand using a five-item scale (Ahluwalia, Burnkrant, and Unnava 2000; Swaminathan, Page, and Gürhan-Canli 2007): "bad/good," "low quality/high quality," "undesirable/desirable," "harmful/beneficial," and "unfavorable/favorable" ($\alpha = .96$). We measured participants' mind perception of agency of the source of the error using Gray, Gray, and Wegner's (2007) seven-item scale, which consisted of the following items: (1) telling right from wrong, (2) remembering things, (3) understanding how others feel, (4) conveying thoughts to others, (5) making plans, (6) exercising self-restraint over impulses, and (7) thinking (1 = "not at all," and 7 = "very much"; $\alpha = .95$). We then measured participants' perceptions of the source of the error's responsibility for the harm caused by the error using Waytz, Heafner, and Epley's (2014) four-item scale. The items are the extent to which the source of the error at Fiat Chrysler (1) was responsible, (2) must be held to account, (3) deserves blame, and (4) was

[4] Following the suggestion of a reviewer, we conducted a pretest (N = 153) that rules out the alternative possibility that people attribute more agency to algorithms because when algorithms make mistakes, people may consider that "even a superior entity that has higher capacities made a mistake." In this pretest, people significantly attributed more agency to humans than they did to algorithms ($M_{human} = 5.95$, SD = .94, $M_{algorithm} = 4.47$, SD = 1.42, $t(152) = 10.409$, $p < .001$).

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blameworthy for the harm caused by the error (1 = "not at all," and 7 = "very much"; $\alpha = .93$).

As a manipulation check, we asked participants to indicate the extent to which they thought the source of the error was a human and the extent to which they thought the source of the error was a computer algorithm (1 = "not at all," and 7 = "very much"). Finally, participants provided their basic demographic information.

Results

Manipulation check. As intended, participants in the human error (vs. algorithm error) condition indicated that the source of the error is more human: $M_{HE} = 5.32$, SD = 1.41 versus $M_{AE} = 4.22$, SD = 1.73, $F(1, 249) = 30.18$, $p < .001$. Participants in the algorithm error (vs. human error) condition indicated that the source of the error is more algorithm-like: $M_{AE} = 5.07$, SD = 1.60 versus $M_{HE} = 4.03$, SD = 1.61, $F(1, 249) = 26.33$, $p < .001$.

Brand attitude. A one-way ANOVA on participants' attitude toward the brand, Fiat Chrysler, is significant: $F(1, 249) = 4.09$, $p = .044$. Supporting H_1, participants' responses to the brand following a brand harm crisis are less negative when the error is an algorithm error (vs. human error): $M_{AE} = 4.59$, SD = 1.61 versus $M_{HE} = 4.17$, SD = 1.69.

Test of mediation. Next, we tested the mediating roles of mind perception of agency of the source of the error and the source of the error's responsibility for the harm caused by the error in mediating participants' responses to the brand following a brand harm crisis (H_2). We note that the means of mind perception of agency and responsibility for the harm caused in the algorithm error and human error conditions are, respectively, as follows: $M_{AE} = 3.65$, SD = 1.65 versus $M_{HE} = 4.87$, SD = 1.37, $F(1, 249) = 40.66$, $p < .001$; $M_{AE} = 4.53$, SD = 1.64 versus $M_{HE} = 5.11$, SD = 1.35, $F(1, 249) = 9.56$, $p = .002$.

We first regressed participants' perceptions of the source of the error's responsibility for the harm caused on algorithm error (vs. human error) condition and found a significant effect: $\beta = .19$, $p = .002$. We then regressed participants' mind perception of the source of the error's agency in committing the error on algorithm error (vs. human error) condition and found a significant effect: $\beta = .37$, $p < .001$. We then regressed participants' perception of the source of the error's responsibility for the harm caused on both algorithm error (vs. human error) condition and mind perception of the source of the error's agency in committing the error. While there is no effect of algorithm error (vs. human error) condition, $\beta = .04$, $p = .54$, there is a significant effect of mind perception of the source of the error's agency in committing the error: $\beta = .41$, $p < .001$.

Next, we formally tested the proposed serial mediation model (H_2). We used PROCESS Macro Model 6 (Hayes and Preacher 2014), where algorithm error (vs. human error) was the independent variable, participants' mind perception of the source of the error's agency in committing the error and perception of the source of the error's responsibility for the harm were the serial mediators, and brand attitude was the dependent variable. The model first tested the effect of the algorithm error (vs. human error) and mind perception of the source of the error's agency in committing the error on perception of the source of the error's responsibility for the harm caused. The results show no effect of algorithm error (vs. human error), $\beta = .1167$, 95% CI = [−.2535, .4869], but a significant effect of mind perception of the source of the error's agency in committing the error on the source of the error's responsibility for the harm caused: $\beta = .3911$, 95% CI = [.2762, .5059].

The model then tested for the effects of algorithm error (vs. human error), mind perception of the source of the error's agency in committing the error, and perception of the source of the error's responsibility for the harm caused on brand attitude. The results show a significant effect of algorithm error (vs. human error) ($\beta = -.6306$, 95% CI = [−1.0555, −.2058]), participants' mind perception of the source of the error's agency in committing the error ($\beta = .3105$, 95% CI = [.1673, .4536]), and perception of the source of the error's responsibility for the harm caused ($\beta = -.2794$, 95% CI = [−.4228, −.1360]) on brand attitude. The 95% bias-corrected bootstrap CI for the indirect effect of algorithm error (vs. human error) on brand attitude is significant ($\beta = -.1309$; 95% CI = [−.2413, −.0498]), indicating serial mediation by mind perception of the source of the error's agency in committing the error and perception of the source of the error's responsibility for the harm caused.

The results of Study 2 offer two findings. First, supporting H_1, consumers' responses to the brand following a brand harm crisis caused by an algorithm (vs. human) error are less negative. Second, in support of H_2, following a brand harm crisis caused by an algorithm error, participants' mind perception of the source of the error's agency in committing the error and perception of the source of the error's responsibility for the harm serially mediate consumers' less negative responses to the brand. As the serial mediation is only partial, there may be other theoretical mechanisms that can be explored in future research.

Study 3: Anthropomorphized Algorithm

In Study 3, we examine H_3, which proposes that consumers' responses to a brand following a brand harm crisis will be more negative when the error is caused by an anthropomorphized (vs. not) algorithm. We use an incentive-compatible experimental design with a consequential outcome, a donation to a Feeding America network suggested by the brand, as the dependent variable. We also measure participants' brand attitude. In this study, a fictitious financial investment company, HMS Investments, is facing a crisis because it made a mistake in its investment decisions for its customers, resulting in financial losses for them. We preregistered this study on AsPredicted.org (#44090).

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Participants and Procedure

A total of 372 adults (180 female, M_{age} = 36.34 years, SD = 14.03) participated in the experiment on the MTurk online platform in exchange for one U.S. dollar. As it is not meaningful to consider anthropomorphized humans, we used a three-factor experimental design consisting of algorithm error, human error, and anthropomorphized algorithm error conditions, to which we randomly assigned participants. We informed participants that HMS Investments, a leading financial investment company, was facing a crisis because a financial algorithm program (financial manager or financial algorithm program Charles) had committed an error, resulting in financial losses for its customers.

We measured participants' attitude toward the brand, HMS Investments, using the same five-item scale used in Study 2 ($\alpha = .97$). We informed participants that the study's researchers decided to randomly give 20 participants a five-dollar bonus, from which participants could donate to Feeding America, the largest domestic hunger-relief organization in the United States through HMS Investments. We informed them that each dollar donated provides about ten meals to families in need through the Feeding America's network of food banks. Participants indicated the amount they were willing to donate to Feeding America, ranging from nothing to five dollars.

As a manipulation check, we asked participants to indicate the extent to which they thought the source of the error at HMS Investments was a human (and algorithm) on a two-item scale (1 = "not at all," and 7 = "very much"). Participants also provided perceptions of the extent to which the news was from a credible source and the extent to which the news was believable on a two-item scale (1 = "not at all," and 7 = "very much"). Results show no effect of algorithm error versus anthropomorphized algorithm error versus human error conditions on the news' credibility, $F(2, 369) = .23$, $p = .794$, or its believability, $F(2, 369) = .653$, $p = .521$. Finally, participants provided their basic demographic information.

Results and Discussion

Manipulation check. As intended, participants in the human error (vs. algorithm error and vs. anthropomorphized algorithm error) condition indicated that the source of the error at HMS Investments was more human: $M_{HE} = 5.84$, SD = 1.24 versus $M_{AE} = 4.15$, SD = 1.95 versus $M_{AAE} = 4.26$, SD = 1.81, $F(2, 369) = 38.48$, $p < .001$. As intended, participants in the algorithm error and anthropomorphized algorithm error (vs. human error) conditions indicated that the source of the error at HMS Investments was more algorithm-like: $M_{AE} = 5.40$, SD = 1.63 versus $M_{AAE} = 5.35$, SD = 1.64 versus $M_{HE} = 3.49$, SD = 1.96, $F(2, 369) = 47.93$, $p < .001$.

Brand attitude. Consistent with H_3, a one-way ANOVA on participants' brand attitude toward HMS Investments is significant: $F(2, 369) = 4.19$, $p = .016$. Supporting H_3, participants' responses to the brand following a brand harm crisis caused by an algorithm error are more negative when the algorithm is anthropomorphized (vs. not): $M_{AAE} = 3.74$, SD = 1.76 versus $M_{AE} = 4.25$, SD = 1.84, $p = .027$. Furthermore, in support of H_1, following a brand harm crisis caused by an algorithm error (vs. human error), participants' brand attitude is less negative, $M_{AE} = 4.25$, SD = 1.84 versus $M_{HE} = 3.63$, SD = 1.81, $p = .008$. In addition, there is no difference in participants' brand attitude when a brand harm crisis is caused by an anthropomorphized algorithm error versus a human error: $M_{AAE} = 3.74$, SD = 1.76 versus $M_{HE} = 3.63$, SD = 1.81, $p = .617$.

Donation amount. Consistent with H_3, a one-way ANOVA on participants' donation is significant: $F(2, 369) = 3.18$, $p = .043$. Participants' donations to Feeding America are higher when the brand harm crisis at HMS Investments is caused by an algorithm error (vs. anthropomorphized algorithm error): $M_{AAE} = 160.40$ (cents), SD = 157.47 versus $M_{AE} = 204.98$ (cents), SD = 180.05, $p = .039$. Following a brand harm crisis, participants' donations to Feeding America are higher when the error is caused by an algorithm error (vs. human error): $M_{AE} = 204.98$ (cents), SD = 180.05 versus $M_{HE} = 156.88$ (cents), SD = 165.19, $p = .029$. Furthermore, there is no difference in participants' donations to Feeding America when the brand harm crisis is caused by an anthropomorphized algorithm error versus a human error: $M_{AAE} = 160.40$ (cents), SD = 157.47 versus $M_{HE} = 156.88$ (cents), SD = 165.19, $p = .864$.

The results of Study 3 offer two key findings. First, supporting H_3, consumers' responses to the brand following a brand harm crisis caused by an algorithm error are more negative when the algorithm is anthropomorphized (vs. not). Second, in support of H_1, consumers' responses to the brand following a brand harm crisis are less negative when the error is caused by an algorithm (vs. human).

Study 4: Machine Learning Algorithm

In Study 4, we examine H_4, which proposes that consumers' responses to a brand following a brand harm crisis will be more negative when the error is caused by a machine learning (vs. not) algorithm. In this study, Twitter was facing a crisis because it made a mistake in the timelines of its users so that some of the displayed tweets had inappropriate and offensive content. We measured participants' attitude toward Twitter following the brand harm crisis.

Participants and Procedure

A total of 310 adults participated in the experiment on the MTurk online platform in exchange for one U.S. dollar (155 male; M_{age} = 34.95 years, SD = 11.04). We informed participants that when users log in to Twitter, their home timelines display a stream of tweets from accounts that they have chosen to follow on Twitter.

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As in Study 3, it was not useful to consider machine learning humans. Therefore, we again used a three-factor experimental design consisting of algorithm error, human error, and machine learning algorithm error conditions to which we randomly assigned participants. In the algorithm error (human error) condition, participants read that Twitter uses algorithms (employees) to evaluate scores and determine which tweets to display. In the machine learning algorithm error condition, participants read that Twitter uses machine learning algorithms to determine which tweets to display. In addition, participants in the machine learning algorithm error condition read that machine learning algorithms are algorithms that learn from past data and analyses to make their decisions. We then informed participants that there had been some problems in Twitter timelines that resulted in the incorrect display of tweets for users. Some of these incorrectly displayed tweets had inappropriate content that had offended some Twitter users.

We measured participants' attitude toward Twitter, using the same five-item scale used in Study 2 ($\alpha = .94$). As a manipulation check, we asked participants for their perceptions of whether the source of the error at Twitter was a human (1 = "not at all," and 7 = "very much"). Participants also provided perceptions of the extent to which the news was from a credible source and the extent to which the news was believable on a two-item scale (1 = "not at all," and 7 = "very much"). Results show no effect of algorithm error versus machine learning algorithm error versus human error conditions on the news' credibility, $F(2, 307) = 1.51$, $p = .22$, or its believability, $F(2, 307) = 1.26$, $p = .28$. Finally, participants indicated whether they had a Twitter account and provided their basic demographic information.

Results and Discussion

Manipulation check. As intended, participants in the human error (vs. algorithm error) condition indicated that the source of the error at Twitter was more human: $M_{HE} = 5.06$, $SD = 1.75$ versus $M_{AE} = 4.49$, $SD = 1.60$, $p = .014$. There is no significant difference between the participants in the human error (vs. machine learning algorithm error) condition regarding the extent to which they thought the source of the error at Twitter was more human: $M_{HE} = 5.06$, $SD = 1.75$ versus $M_{MLAE} = 4.79$, $SD = 1.54$, $p = .243$. There is also no significant difference between participants in the algorithm error (vs. machine learning algorithm error) condition regarding the extent to which they thought the source of the error at Twitter was more human: $M_{AE} = 4.49$, $SD = 1.60$ versus $M_{MLAE} = 4.79$, $SD = 1.54$, $p = .176$.

Brand attitude. Consistent with H_4, a one-way ANOVA on participants' attitude toward Twitter is significant, $F(2, 307) = 4.72$, $p = .010$. Supporting H_4, participants' responses to the brand following a brand harm crisis are more negative when the error was caused by a machine learning (vs. not) algorithm: $M_{MLAE} = 4.21$, $SD = 1.45$ versus $M_{AE} = 4.76$, $SD = 1.62$, $p = .011$. Furthermore, in support of H_1, following a brand harm crisis caused by an algorithm error (vs. human error), participants' responses to the brand are less negative: $M_{AE} = 4.76$, $SD = 1.62$ versus $M_{HE} = 4.20$, $SD = 1.47$, $p = .009$. There is no difference in participants' responses to the brand following a brand harm crisis caused by a machine learning algorithm versus a human: $M_{MLAE} = 4.21$, $SD = 1.45$ versus $M_{HE} = 4.20$, $SD = 1.47$, $p = .95$. Whether participants have a Twitter account did not change the effect of the algorithm error versus machine learning algorithm error versus human error condition on their attitude toward Twitter, $F(2, 306) = 4.58$, $p = .011$.

The results of Study 4 offer two key findings. First, supporting H_4, consumers' responses to the brand following a brand harm crisis are more negative when the error is caused by a machine learning (vs. not) algorithm. Second, in support of H_1, following a brand harm crisis caused by an algorithm error (vs. human error), consumers' responses to the brand are less negative.

Study 5: Subjective (vs. Objective) Task

In Study 5, we test H_5, which proposes that consumers' responses to a brand following a brand harm crisis will be more negative when the algorithm error occurs during a subjective (vs. objective) task. In this study, we informed participants that a leading university in the United States was facing a crisis because of an error in the subjective (vs. objective) assessment of Asian American students' applications. We measured participants' attitude toward the university. We preregistered the study at AsPredicted.org (#43396).

Participants and Procedure

A total of 400 adults (199 female, $M_{age} = 35.70$ years, $SD = 11.97$) participated in the study on the MTurk online platform in exchange for monetary compensation. We used a 2 (algorithm error, human error) × 2 (subjective task, objective task) between-subjects design.

We informed participants that a leading university in the United States was experiencing a crisis because of a mistake in assessments of the applications of prospective Asian American students. We randomly assigned participants to either the algorithm error or human error condition. In the algorithm error (human error) condition, we informed participants that the computer algorithm (employees) had made the mistake.

We informed participants that the university used both subjective and objective methods in their admissions process. The subjective methods included analyzing the applicant's personality and social skills, including "positive personality," likability, courage, kindness, and being "widely respected." The objective methods included reviewing the applicant's test scores and grades. Participants in the subjective (objective) task condition read that the error was in the subjective (objective) assessment of the application. Participants in the subjective task condition read that Asian American applicants were rated lower than other applicants on traits like positive

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personality, likability, courage, kindness, and being widely respected. Participants in the objective task condition read that the error was based on the incorrect use of lower test scores for the Asian American applicants. In both conditions, participants read that the error resulted in the university incorrectly declining applications of hundreds of qualified and acceptance-worthy Asian American applicants.

We measured participants' attitude toward the university using the same five-item scale used in Study 2 ($\alpha = .95$). As a manipulation check, we asked participants to indicate the extent to which they thought the source of the error was human, the extent to which they thought the source of the error was an algorithm, and the extent to which they thought the error was related to an objective task (1 = "not at all," and 7 = "very much"). Participants also provided perceptions of the extent to which the news was from a credible source and the extent to which the news was believable (1 = "not at all," and 7 = "very much"). Results showed no effect of algorithm error versus human error conditions and subjective versus objective task conditions on the news' credibility, $F(1, 396) = .00$, $p = .994$, or its believability, $F(1, 396) = .454$, $p = .501$. Finally, participants provided their basic demographic information.

Results and Discussion

Manipulation check. As intended, participants in the human error (vs. algorithm error) condition indicated that the source of the error was more human: $M_{HE} = 5.33$, $SD = 1.41$ versus $M_{AE} = 4.45$, $SD = 1.51$, $t(398) = 6.003$, $p < .001$. Participants in the algorithm error (vs. human error) condition indicated that the source of the error was more algorithm-like: $M_{AE} = 4.68$, $SD = 1.64$ versus $M_{HE} = 3.84$, $SD = 1.68$, $t(398) = -5.047$, $p < .001$. Participants in the objective task (vs. subjective task) condition indicated that the error was likely to have occurred during an objective task: $M_{objective} = 4.60$, $SD = 1.41$ versus $M_{subjective} = 3.94$, $SD = 1.74$, $t(398) = -4.184$, $p < .001$.

Brand attitude. Consistent with H_5, an ANOVA on participants' brand attitude reveals the predicted interaction effect of algorithm error (vs. human error) and subjective (vs. objective) task conditions, $F(1, 396) = 9.15$, $p = .003$. Supporting H_5, participants' responses to a brand following a brand harm crisis caused by an algorithm error are more negative when the error occurs during a subjective (vs. objective) task: $M_{subjective} = 3.76$, $SD = 1.64$ versus $M_{objective} = 4.46$, $SD = 1.49$, $F(1, 396) = 9.06$, $p = .003$. There is no difference in participants' responses to a brand following a brand harm crisis caused by a human error when the error occurs during a subjective versus objective task: $M_{subjective} = 4.28$, $SD = 1.73$ versus $M_{objective} = 3.99$, $SD = 1.64$, $F(1, 396) = 1.58$, $p = .21$.

In support of H_1, participants' responses to a brand following a brand harm crisis caused by an algorithm error (vs. human error) are less negative when the error occurs during an objective task: $M_{AE} = 4.46$, $SD = 1.49$ versus $M_{HE} = 3.99$, $SD = 1.64$, $F(1, 396) = 4.13$, $p = .043$. Participants' responses to a brand following a brand harm crisis caused by an algorithm error (vs. human error) are more negative when the error occurs during a subjective task: $M_{AE} = 3.76$, $SD = 1.64$ versus $M_{HE} = 4.28$, $SD = 1.73$, $F(1, 396) = 5.04$, $p = .025$.

The results of Study 5 offer two key findings. First, supporting H_5, consumers' responses to a brand following a brand harm crisis are more negative when the algorithm error occurs during a subjective (vs. objective) task. Second, supporting H_1, consumers' responses to the brand following a brand harm crisis caused by an algorithm error (vs. human error) are less negative when the error occurs during an objective task.

Study 6: Interactive Task

In Study 6, we test H_6, which proposes that consumers' responses to a brand following a brand harm crisis caused by an algorithm error will be more negative when there is interactivity (vs. not) with the algorithm during the task in which the error occurs. We informed participants that a fictitious leading fashion retailer brand, D&J, has been facing growing customer complaints because of its personal stylists. Participants were randomly assigned to algorithm error (vs. human error) and interactive (vs. noninteractive) task conditions. We measured participants' brand attitude.

Participants and Procedure

A total of 328 students (206 female, $M_{age} = 20.12$ years, $SD = 1.64$) from a university in the southern United States participated in the laboratory experiment in exchange for course credit. We used a 2 (algorithm error, human error) × 2 (interactive task: yes, no) between-subjects design.

We randomly assigned participants to either the algorithm error or human error conditions. Participants in the algorithm error (human error) condition read that in recent weeks, D&J, a leading fashion retailer brand, had been facing growing customer complaints because of problems caused by its algorithm (human) personal stylists, which were recently introduced by D&J and were intended to personalize products for customers to reflect and accentuate their personalities. Participants were assigned to the interactive and noninteractive task conditions. To ensure realism, we did not use the word "human" in the human error condition.

Participants in the interactive task condition read that customers who wanted to use the interactive algorithm (personal) stylists first completed an online form, which collected a personal photograph and details of customers' height, weight, and personal likes and dislikes of different colors and styles. Then, the D&J algorithm (personal) stylists interacted with customers such that customers could see how the products would look on them and could work with the D&J algorithm (personal) stylists to choose the right products. The customer was thus actively involved in the selection of products by algorithm (personal) stylists. Using the information provided by the customer, the algorithm (personal) stylists chose and shipped products to customers.

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Participants in the noninteractive task condition read that customers who used the algorithm (personal) stylists first completed an online form that collected a personal photograph and details of their height, weight, and personal likes and dislikes of different colors and styles. Then, the D&J algorithm (personal) stylists chose the right products for the customer. The customer was not involved in the selection of products done by algorithm (personal) stylists. Using the information provided by the customer, the algorithm (personal) stylists chose and shipped products to customers. Participants read that customers stated stylists misled them, causing them to buy very expensive products that did not reflect their personalities and were, in fact, mismatched with their personalities. Customers were now demanding refunds for these products and threatening to sue D&J.

We measured participants' brand attitude, using the same five-item scale used in Study 2 ($\alpha = .88$). As a manipulation check, we asked participants to indicate the extent to which they thought the source of the error at D&J was a human and the extent to which they thought the error occurred during a task in which there was communication between the personal stylist and the customer, which indicates interactivity on a two-item scale (1 = "not at all," and 7 = "very much"). Participants also provided perceptions of the extent to which the news was from a credible source and the extent to which the news was believable on a two-item scale (1 = "not at all," and 7 = "very much"). Results showed no effect of algorithm error versus human error condition and interactive versus noninteractive task conditions on the news' credibility, $F(1, 324) = 1.49$, $p = .22$, or its believability, $F(1, 324) = .018$, $p = .89$. Finally, participants provided their basic demographic information.

Results and Discussion

Manipulation check. As intended, participants in the human error (vs. algorithm error) condition indicated that the source of the error was more human: $M_{HE} = 5.19$, $SD = 1.47$ versus $M_{AE} = 4.40$, $SD = 1.51$, $t(326) = 4.83$, $p < .001$. Participants in the interactive (vs. noninteractive) task indicated that the error was more likely to have occurred on a task in which there was more communication between the personal stylist and the customer: $M_{interactive} = 3.51$, $SD = 1.57$ versus $M_{noninteractive} = 3.04$, $SD = 1.41$, $t(326) = -2.90$, $p = .004$.

Brand attitude. Consistent with H_6, an ANOVA on brand attitude reveals the predicted interaction effect of algorithm error (vs. human error) and interactive (vs. noninteractive) task conditions, $F(1, 324) = 5.05$, $p = .025$. Supporting H_6, participants' responses to a brand following a brand harm crisis caused by an algorithm error are more negative when there is interactivity (vs. not) with the algorithm during the task in which the error occurs: $M_{interactive} = 3.41$, $SD = 1.05$ versus $M_{not} = 3.82$, $SD = 1.22$, $F(1, 324) = 6.33$, $p = .012$. There is no difference in participants' responses to a brand following a brand harm crisis caused by a human error when there was interactivity versus no interactivity with the employee during the task: $M_{interactive} = 3.48$, $SD = .98$ versus $M_{noninteractive} = 3.37$, $SD = .95$, $F(1, 324) = .44$, $p = .51$.

In support of H_1, participants' responses to a brand following a brand harm crisis caused by an algorithm error (vs. human error) are less negative when the task in which the error occurs is noninteractive: $M_{AE} = 3.82$, $SD = 1.22$ versus $M_{HE} = 3.37$, $SD = .95$, $F(1, 324) = 7.59$, $p = .006$. There is no difference in participants' responses to a brand following a brand harm crisis caused by an algorithm error versus a human error when the task was interactive: $M_{AE} = 3.41$, $SD = 1.05$ versus $M_{HE} = 3.48$, $SD = .98$, $F(1, 324) = .179$, $p = .672$.

The results of Study 6 offer two findings. First, supporting H_6, consumers' responses to a brand following a brand harm crisis caused by an algorithm error are more negative when the task in which the error occurs is interactive (vs. noninteractive). Second, supporting H_1, consumers' responses to the brand following a brand harm crisis caused by an algorithm (vs. human) error are less negative when the task in which the error occurs is noninteractive.

Managerial Study M1

As algorithm errors are, unfortunately, common in business practice, firms undertake interventions to manage the aftermath of such brand harm crises. The baseline intervention in algorithm errors is technological supervision of the algorithm (e.g., facial recognition algorithm failures at Microsoft; Roach 2018). Another common intervention following brand harm crises caused by an algorithm error is to increase human supervision of the algorithm (Lee, Resnick, and Barton 2019). As Sheryl Sandberg, Chief Operating Officer, Facebook, noted (in 2017) after an algorithm error caused the display of anti-Semitic ads, "we're adding more human review and oversight to our automated processes. . . . From now on we will have more manual review of new ad targeting options to help prevent offensive terms from appearing." To generate managerial guidance, we conducted a study (M1) in which we examined consumers' responses to human supervision and technological supervision following brand harm crises caused by an algorithm (vs. human) error.

Participants and Procedure

A total of 368 adults (171 female, $M_{age} = 35.08$ years, $SD = 11.06$) participated in the study on the MTurk online platform in exchange for monetary compensation. We used a 2 (algorithm error, human error) × 2 (human supervision, technological supervision) between-subjects design. We preregistered the study at AsPredicted.org (#53178).

All participants saw a tweet on the official Twitter account of *The New York Times* announcing the recall of 4.8 million Fiat Chrysler vehicles due to a cruise control problem. We assigned participants to the algorithm error and human error conditions. Participants in the algorithm error condition read that the computer algorithm at Fiat Chrysler had made a mistake resulting in a defect in the cruise control system, causing a

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safety hazard. Participants in the human error condition read that the employees of Fiat Chrysler had made a mistake resulting in a defect in the cruise control system, causing a safety hazard. We then randomly assigned participants to human supervision and technological supervision conditions. We informed participants in the human supervision condition that Fiat Chrysler would increase managerial supervision in its manufacturing processes to prevent such errors. We informed participants in the technological supervision condition that Fiat Chrysler would increase technological supervision in the manufacturing processes to prevent such errors.

We measured participants' attitude toward the Fiat Chrysler brand using the same five-item scale used in Study 2 ($\alpha = .96$). As a manipulation check, we asked participants to indicate the extent to which they thought the source of the error was human, the extent to which they thought the source of the error was an algorithm, the extent to which they thought there will be more human supervision at Fiat Chrysler in the future, and the extent to which there will be more technological supervision at Fiat Chrysler in the future on four seven-point scales (1 = "not at all," and 7 = "very much"). Participants also provided perceptions of the extent to which the news was believable (1 = "not at all," and 7 = "very much"). Results show no effect of algorithm error versus human error conditions and human supervision versus technological supervision conditions on the news' believability, $F(1, 364) = .152$, $p = .697$. Finally, participants provided their basic demographic information.

Results

Manipulation check. As intended, participants in the human error (vs. algorithm error) condition indicated that the source of the error was more human: $M_{HE} = 5.12$, $SD = 1.54$ versus $M_{AE} = 4.54$, $SD = 1.75$, $t(366) = -3.41$, $p = .001$. Participants in the algorithm error (vs. human error) condition indicated that the source of the error was more algorithm-like: $M_{HE} = 3.93$, $SD = 1.64$ versus $M_{AE} = 4.92$, $SD = 1.57$, $t(366) = 5.90$, $p < .001$. Participants in the human supervision (vs. technological supervision) condition indicated, going forward, there will be more human supervision at Fiat Chrysler: $M_{HS} = 5.41$, $SD = 1.44$ versus $M_{TS} = 5.06$, $SD = 1.47$, $t(366) = 2.28$, $p = .023$. Participants in the technological supervision (vs. human supervision) condition indicated, going forward, there will be more technological supervision at Fiat Chrysler: $M_{HS} = 4.95$, $SD = 1.67$ versus $M_{TS} = 5.29$, $SD = 1.40$, $t(366) = -2.09$, $p = .037$.

Brand attitude. An ANOVA on brand attitude reveals the predicted interaction effect of algorithm error (vs. human error) and human supervision (vs. technological supervision) conditions: $F(1, 364) = 9.25$, $p = .003$. Participants' responses to a brand following a brand harm crisis caused by an algorithm error are more negative when there is more human supervision (vs. technological supervision): $M_{HS} = 4.13$, $SD = 1.56$ versus $M_{TS} = 4.71$, $SD = 1.69$, $F(1, 364) = 5.44$, $p = .020$. Participants' responses to a brand following a brand harm crisis caused by a human error are marginally more negative when there is more technological supervision (vs. human supervision): $M_{HS} = 4.63$, $SD = 1.69$ versus $M_{TS} = 4.15$, $SD = 1.65$, $F(1, 364) = 3.87$, $p = .050$.

Participants' responses to a brand following a brand harm crisis caused by an algorithm error (vs. human error) are less negative when there is more technological supervision: $M_{AE} = 4.71$, $SD = 1.69$ versus $M_{HE} = 4.15$, $SD = 1.65$, $F(1, 364) = 5.27$, $p = .022$. Participants' responses to a brand following a brand harm crisis caused by an algorithm error (vs. human error) are more negative when there is more human supervision: $M_{AE} = 4.13$, $SD = 1.56$ versus $M_{HE} = 4.63$, $SD = 1.69$, $F(1, 364) = 4.02$, $p = .046$.

Study M1's findings indicate that consumers' responses to a brand following a brand harm crisis caused by an algorithm error are more (less) negative when there is human (technological) supervision of the algorithm following the harm crisis. The practical implication of these findings is that marketers should not (should) publicize human (technological) supervision of algorithms in communications with customers to ensure the best possible responses from consumers when the brand introduces such supervision following a brand harm crisis caused by an algorithm error.

General Discussion

> "AI algorithms may be flawed . . . These deficiencies could undermine the decisions, predictions, or analysis AI applications produce, subjecting us to competitive harm, legal liability, and brand or reputational harm."
>
> Microsoft Annual Report 2018.

The use of algorithmic marketing across many applications is growing dramatically in many sectors. Moreover, there is growing evidence of the occurrence of algorithm errors that cause brand harm crises. However, there are few insights in the marketing literature on consumers' responses to brands following brand harm crises caused by algorithm errors.

Addressing this research gap, we develop and find support for a theory of consumers' responses to a brand following a brand harm crisis caused by an algorithm error. The findings from eight experimental studies that support our hypotheses are robust across multiple contexts (e.g., products, financial services, and online services), different samples (e.g., students and adults), and different responses, including attitudinal, behavioral, and consequential actions (in two incentive-compatible experimental designs). We conclude with a discussion of the findings' theoretical contributions, managerial implications, and limitations and opportunities for further research.

Theoretical Contributions

Harm crises. Distinct from past research on consumers attributing product failures to managerial (i.e., human) errors, we consider brand harm crises caused by inanimate entities:

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algorithms that are software programs. Consumers perceive that inanimate algorithms have lower agency over the error and therefore lower responsibility for the harm caused by the error.

Applying the theory of mind perception (Gray, Gray, and Wegner 2007) to algorithms committing errors that cause brand harm crises, we find that consumers have lower mind perception of agency of the algorithm for the error and assign lower responsibility to the algorithm for the harm caused (H_2), resulting in less negative responses to the brand (H_1). Furthermore, consumers' responses to the brand following a brand harm crisis caused by an algorithm error are more negative when (1) the algorithm is anthropomorphized (vs. not) (H_3), (2) the algorithm is a machine learning algorithm (vs. not) (H_4), (3) the algorithm error occurs during a subjective (vs. objective) task (H_5), and (4) the algorithm error occurs during an interactive (vs. noninteractive) task (H_6).

Taken together, the support for the four moderation effects (i.e., whether the algorithm is anthropomorphized, whether the algorithm involves machine learning, whether the task is subjective, and whether the task is interactive), each of which humanize the algorithm, indirectly support the serial mediation by lower mind perception of agency of the algorithm for the error and, in turn, consumers' lower perception of the algorithm's responsibility for the harm caused. Given the growing prevalence of inanimate entities (e.g., algorithms, robots, drones) in practice, this research's findings make a novel contribution to the literature on harm crises, which has not examined consumers' responses to errors caused by inanimate entities. Furthermore, extant literature has also not examined moderators of the sources of harm crises and characteristics of the task in which the error occurs. Finally, the support for partial serial mediation by agency of the algorithm and, in turn, algorithms' lower responsibility for the harm caused suggests there may be other theoretical processes that offer future research opportunities.

Algorithm usage. Extant research on algorithm usage (e.g., Dietvorst, Simmons, and Massey 2015; Logg, Minson, and Moore 2019; Prahl and Van Swol 2017) has focused on consumers' decisions to use (or continue to use) an algorithm. However, there may be situations in practice, such as in algorithmic marketing, in which others, not the algorithm users, decide whether to deploy the algorithm. Algorithm errors frequently occur even in these contexts, which is an issue overlooked in extant research. We address this gap and consider consumers' responses to the brand following a brand harm crisis caused by an algorithm error (vs. human error) in which brand managers (not consumers) decide to deploy the algorithm. In what we consider a novel finding, when an algorithm commits an error and causes a brand harm crisis, consumers' responses to the brand following the crisis are less negative than if the firm's managers committed the same error. That is, consumers are more forgiving of algorithm errors, which suggests individuals' receptivity to algorithms when they do not have the decision-making authority on whether to use the algorithm.

Further research on individuals' responses to algorithm errors will be useful, for example, in health care, in which there is increasing application of algorithms where users have no say in their usage. For example, in the diagnosis and treatment of health conditions for which big data are used, different types of errors may occur (e.g., omission or commission, Type I and Type II errors resulting in false positives and false negatives) that could affect consumers' responses to the brand using the algorithm and to the algorithm itself.

Algorithmic marketing. To the best of our knowledge, this is the first study to examine consumers' responses to algorithmic errors. We identify consumers' mind perception of agency of algorithms as a building block that is relevant in the study of algorithmic marketing. Moreover, drawing on the findings of the four moderation effects, we identify conditions related to error source and task characteristics that modify the main effect of the algorithm error on consumers' responses to the brand. In doing so, we identify building blocks for developing a comprehensive theory of algorithmic marketing. Relevant questions for further research include how consumers may respond to the brand across different algorithm errors in product development, advertising, and targeting settings. A research area with policy implications is the ethicality of algorithmic marketing (e.g., facial recognition algorithms inappropriately targeting/excluding minorities) (Spirina 2009).

Managerial Implications

Our findings offer actionable guidance to managers on the deployment of algorithms in marketing contexts. First, consumers' responses to a brand following a brand harm crisis caused by an algorithm error (vs. human error) are less negative. In addition, consumers' perceptions of the algorithm's lower agency for the error and resultant lower responsibility for the harm caused mediate their responses to a brand following a brand harm crisis caused by an algorithm error. In summary, consumers penalize brands less when an algorithm (vs. human) causes an error that leads to a brand harm crisis.

Second, the findings show that consumers' responses to a brand are more negative following a brand harm crisis caused by an algorithm error when the algorithm appears more human. Thus, a brand's risk exposure to the harm caused by an algorithm error is higher when the algorithm is anthropomorphized (vs. not), involves machine learning (vs. not), is used in a subjective (vs. objective) task, or is used in an interactive (vs. noninteractive) task. Marketers must be aware that in contexts in which an algorithm appears to be more human, it would be wise to have heightened vigilance in its deployment and monitoring and to allocate resources for managing the aftermath of any brand harm crises caused by algorithm errors.

Third, to manage the aftermath of brand harm crises caused by algorithm errors, managers can highlight the role of the algorithm and the lack of agency of the algorithm for the error, which may attenuate consumers' negative responses to the brand. However, we caution that highlighting the role of the

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algorithm will worsen the situation by strengthening consumers' negative responses if the error involves an anthropomorphized algorithm or a machine learning algorithm, or if the algorithm error occurs in a subjective or interactive task.

Fourth, the insights from Study M1 generate concrete guidance for effectively managing the aftermath of brand harm crises caused by algorithm errors. Marketers should not publicize human supervision of algorithms (which may actually be effective in fixing the algorithm) in communications with their customers following brand harm crises caused by algorithm errors. However, if a brand uses technological supervision of the algorithm to respond to an error, marketers should publicize this fact to leverage the benefit identified in Study M1 (i.e., that consumers are less negative when there is technological supervision of the algorithm following a brand harm crisis).

Limitations and Further Research

First, in this initial study on brand harm crises caused by algorithm errors, we focus on consumers' negative responses to brands following one algorithm error. We do not consider the effects of repeated algorithm errors. We also do not consider very serious harm crises with dozens of fatalities (e.g., the Lion Air plane crash in Indonesia in October 2018 and the Ethiopian Airlines plane crash in March 2019 caused by Boeing 737 Max 8 automated flight system algorithms). In such cases, we anticipate extremely negative responses in both algorithm and human error conditions, precluding lab experiments for theory testing. Furthermore, we do not consider marketing mix remedies (e.g., advertising, promotions) that may be effective in handling the aftermath of brand harm crises. Further research on brand harm crises caused by algorithm errors in which brands incorporate marketing mix remedies and the effects on brand performance using less intrusive, qualitative methods, including observational studies, would be useful. Second, we focus only on errors in algorithmic marketing. Additional research on harm crises caused by algorithmic errors in other contexts (e.g., health care, justice) in which algorithm usage is increasing and errors have substantive consequences with policy implications would be useful. Third, with respect to the various parties involved, we consider the brand as the focus of our research without consideration of whether there is a distinction between blaming the algorithm itself, the person who designed it, or the person/company that chose to implement it. Whether consumers differentiate between the brand, the designer, or the person implementing the algorithm (e.g., brand manager) is a future research opportunity.

In summary, we view this study as a useful first step in exploring algorithmic marketing by focusing on brand harm crises caused by algorithm errors that, unfortunately, are now rather common in marketing practice. We hope that this research stimulates further work on algorithmic marketing strategies and related consumer behaviors.

Associate Editor

Don Lehmann

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

References

Aggarwal, Pankaj and Ann L. McGill (2007), "Is That Car Smiling at Me? Schema Congruity as a Basis for Evaluating Anthropomorphized Products," *Journal of Consumer Research*, 34 (4), 468–79.

Aggarwal, Pankaj and Ann L. McGill (2012), "When Brands Seem Human, Do Humans Act Like Brands? Automatic Behavioral Priming Effects of Brand Anthropomorphism," *Journal of Consumer Research*, 39 (2), 307–23.

Ahluwalia, Rohini, Robert E. Burnkrant, and H. Rao Unnava (2000), "Consumer Response to Negative Publicity: The Moderating Role of Commitment," *Journal of Marketing Research*, 37 (2), 203–14.

Awad, Edmond, Sydney Levine, Max Kleiman-Weiner, Sohan Dsouza, Joshua B. Tenenbaum, Azim Shariff, et al. (2020), "Drivers are Blamed More Than Their Automated Cars When Both Make Mistakes," *Nature Human Behaviour*, 4 (2), 34–143.

Badger, Emily (2019), "Who's to Blame When Algorithms Discriminate?" *The New York Times* (August 20), https://www.nytimes.com/2019/08/20/upshot/housing-discrimination-algorithms-hud.html.

Castelo, Noah, Maarten W. Bos, and Donald R. Lehmann (2019), "Task-Dependent Algorithm Aversion," *Journal of Marketing Research*, 56 (5), 809–25.

Choi, Sungwoo, Anna S. Mattila, and Lisa E. Bolton (2020), "To Err Is Human(-oid): How Do Consumers React to Robot Service Failure and Recovery?" *Journal of Service Research* (published online December 16), DOI:10.1177/1094670520978798.

Cleeren, Kathleen, Marnik G. Dekimpe, and Harald van Heerde (2017), "Marketing Research on Product-Harm Crises: A Review, Managerial Implications, and an Agenda for Future Research," *Journal of the Academy of Marketing Science*, 45 (5), 593–615.

Diakopoulos, Nicholas (2013), "Race Against the Algorithms," *The Atlantic* (October 3) https://www.theatlantic.com/technology/archive/2013/10/rage-against-the-algorithms/280255/.

Dietvorst, Berkeley, Joseph P. Simmons, and Cade Massey (2015), "Algorithm Aversion: People Erroneously Avoid Algorithms After Seeing Them Err," *Journal of Experimental Psychology: General*, 144 (1), 114–26.

Dutta, Sujay and Chris Pullig (2011), "Effectiveness of Corporate Responses to Brand Crises: The Role of Crisis Type and Response Strategies," *Journal of Business Research*, 64 (12), 1281–87.

Epley, Nicholas, Eugene Caruso, and Max H. Bazerman (2006), "When Perspective Taking Increases Taking: Reactive Egoism in Social Interaction," *Journal of Personality and Social Psychology*, 91 (5), 872–89.

Epley, Nicholas and Adam Waytz (2009), "Mind Perception," in *The Handbook of Social Psychology*, 5th ed, Susan T. Fiske, Daniel T.

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Gilbert and Gardener Lindzey, eds. New York, NY: Wiley, 498–541.

Fletcher, Joseph F. (1979), *Humanhood: Essays in Biomedical Ethics*. Buffalo, NY: Prometheus Books.

Folkes, Valerie S. (1984), "Consumer Reactions to Product Failure: An Attributional Approach," *Journal of Consumer Research*, 10 (4), 398–409.

Folkes, Valerie S. (1990). "Conflict in the Marketplace: Explaining Why Products Fail," in *Attribution Theory: Applications to Achievement, Mental Health, and Interpersonal Conflict*, Sandra Graham and Valerie S. Folkes, eds. Hillsdale, NJ: Lawrence Erlbaum, 143–60.

Folkes, Valerie S., Susan Koletsky, and John L. Graham (1987), "A Field Study of Causal Inferences and Consumer Reaction: The View from the Airport," *Journal of Consumer Research*, 13 (4), 534–39.

Gal, Michal S. and Niva Elkin-Koren (2017), "Algorithmic Consumers," *Harvard Journal of Law & Technology*, 30 (2), 309–53.

Gill, Tripat (2020), "Blame It on the Self-Driving Car: How Autonomous Vehicles Can Alter Consumer Morality," *Journal of Consumer Research*, 47 (2), 272–91.

Gray, Heather M., Kurt Gray, and Daniel M. Wegner (2007), "Dimensions of Mind Perception," *Science*, 315 (5812), 619.

Gray, Kurt and Daniel M. Wegner (2009), "Moral Typecasting: Divergent Perceptions of Moral Agents and Moral Patients," *Journal of Personality and Social Psychology*, 96 (3), 505–20.

Gray, Kurt and Daniel M. Wegner (2012), "Feeling Robots and Human Zombies: Mind Perception and the Uncanny Valley," *Cognition*, 125 (1), 125–30.

Gray, Kurt, Liane Young, and Adam Waytz (2012), "Mind Perception Is the Essence of Morality," *Psychological Inquiry*, 23 (2), 101–24.

Griffith, Eric (2017), "10 Embarrassing Algorithm Fails," *PCMag* (September 23), https://www.pcmag.com/feature/356387/10-embarrassing-algorithm-fails.

Hayes, Andrew F. and Kristopher J. Preacher (2014), "Statistical Mediation Analysis with a Multicategorical Independent Variable," *British Journal of Mathematical and Statistical Psychology*, 67 (3), 451–70.

Heller, Martin (2019), "Machine Learning Algorithms Explained," *InfoWorld* (May 9), https://www.infoworld.com/article/3394399/machine-learning-algorithms-explained.html.

Inbar, Yoel, Jeremy Cone, and Thomas Gilovich (2010), "People's Intuitions About Intuitive Insight and Intuitive Choice," *Journal of Personality and Social Psychology*, 99 (2), 232–47.

Kim, Hyeongmin Christian and Thomas Kramer (2015), "Do Materialists Prefer the 'Brand-as-Servant?' The Interactive Effect of Anthropomorphized Brand Roles and Materialism on Consumer Responses," *Journal of Consumer Research*, 42 (2), 284–99.

Kim, Sara and Ann McGill (2011), "Gaming with Mr. Slot or Gaming the Slot Machine? Power, Anthropomorphism, and Risk Perception," *Journal of Consumer Research*, 38 (1), 94–107.

Kleinberg, Jon, Himabindu Lakkaraju, Jure Leskovec, Jens Ludwig, and Sendhil Mullainathan (2018), "Human Decisions and Machine Predictions," *The Quarterly Journal of Economics*, 133 (1), 237–93.

Lafrance, Adrienne (2014), "Why People Name Their Machines," *The Atlantic* (June 23), https://www.theatlantic.com/technology/archive/2014/06/why-people-give-human-names-to-machines/373219/.

Landwehr, Jan R., Ann McGill, and Andreas Herrmann (2011), "It's Got the Look: The Effect of Friendly and Aggressive 'Facial' Expressions on Product Liking and Sales," *Journal of Marketing*, 75 (3), 132–46.

Lee, Nicol Turner, Paul Resnick, and Genie Barton (2019), "Algorithmic Bias Detection and Mitigation: Best Practices and Policies to Reduce Consumer Harms," *Brookings* (May 22), https://www.brookings.edu/research/algorithmic-bias-detection-and-mitigation-best-practices-and-policies-to-reduce-consumer-harms/.

Lei, Jing, Niraj Dawar, and Zeynep Gürhan-Canli (2012), "Base-Rate Information in Consumer Attributions of Product-Harm Crises," *Journal of Marketing Research*, 49 (3), 336–48.

Logg, Jennifer, Julia Minson, and Don A. Moore, (2019), "Algorithm Appreciation: People Prefer Algorithmic to Human Judgment," *Organizational Behavior and Human Decision Processes*, 151, 90–103.

McCullom, Rod (2017), "Facial Recognition Technology Is Both Biased and Understudied," Undark (May 17), https://undark.org/article/facial-recognition-technology-biased-understudied/.

Microsoft Annual Report (2018), United States Securities and Exchange Commission Form 10-K (accessed June 1, 2021), https://www.sec.gov/Archives/edgar/data/789019/000156459018019062/msft-10k_20180630.htm.

Moon, Youngme (2000), "Intimate Exchanges: Using Computers to Elicit Self-Disclosure from Consumers," *Journal of Consumer Research*, 26 (4), 323–39.

Moon, Youngme (2003), "Don't Blame the Computer: When Self-Disclosure Moderates the Self-Serving Bias," *Journal of Consumer Psychology*, 13 (1/2), 125–37.

Nass, Clifford and Youngme Moon (2000), "Machines and Mindlessness: Social Responses to Computers," *Journal of Social Issues*, 56 (1), 81–103.

Prahl, Andrew and Lyn van Swol (2017), "Understanding Algorithm Aversion: When Is Advice from Automation Discounted?" *Journal of Forecasting*, 36 (6), 691–702.

Pullig, Chris, Richard G. Netemeyer, and Abhijit Biswas (2006), "Attitude Basis, Certainty, and Challenge Alignment: A Case of Negative Brand Publicity," *Journal of the Academy of Marketing Science*, 34 (4), 528–42.

Puzakova, Marina, Hyokjin Kwak, and Joseph F. Rocereto (2013), "When Humanizing Brands Goes Wrong: The Detrimental Effect of Brand Anthropomorphization Amid Product Wrongdoings," *Journal of Marketing*, 77 (3), 81–100.

Rafaeli, Sheizaf (1988), "Interactivity: From New Media to Communication," in *SAGE Annual Review of Communication Research: Advancing Communication Science*, Vol. 16, Robert P. Hawkins, John M. Wiemann and Suzanne Pingree, eds. Beverly Hills, CA: SAGE Publications, 110–34.

Roach, John (2018), "Microsoft Improves Facial Recognition Technology to Perform Well Across All Skin Tones, Genders," *Microsoft: The AI Blog* (June 26), https://blogs.microsoft.com/ai/gender-skin-tone-facial-recognition-improvement/.

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Sandberg, Sheryl (2017), Facebook Post (September 20), https://www.facebook.com/sheryl/posts/10159255449515177.

Spirina, Katrine (2009), "Ethics of Facial Recognition: How to Make Business Uses Fair and Transparent," *Towards Data Science* (April 2019), https://towardsdatascience.com/ethics-of-facial-recognition-how-to-make-business-uses-fair-and-transparent-98e3878db08d.

Sundar, S. Shyam (2009), "Media Effects 2.0: Social and Psychological Effects of Communication Technologies," In *The SAGE Handbook of Media Processes and Effects*, Robin L. Nabi and Mary Beth Oliver, eds. Thousand Oaks, CA: SAGE Publications, 545–60.

Sundar, S. Shyam, Saraswathi Bellur, Jeeyun Oh, Haiyan Jia, and Hyang Sook Kim (2016), "Theoretical Importance of Contingency in Human–Computer Interaction: Effects of Message Interactivity on User Engagement," *Communication Research*, 43 (5), 595–625.

Sundar, S. Shyam, Eun Go, Hyang-Sook Kim, and Bo Zhang (2015), "Communicating Art, Virtually! Psychological Effects of Technological Affordances in a Virtual Museum," *International Journal of Human–Computer Interaction*, 31 (6), 385–401.

Swaminathan, Vanitha, Karen L. Page, and Zeynep Gürhan-Canli (2007), "'My' Brand or 'Our' Brand: The Effects of Brand Relationship Dimensions and Self-Construal on Brand Evaluations," *Journal of Consumer Research*, 34 (2), 248–59.

Sweeney, Latanya (2013), "Discrimination in Online Ad Delivery," *Queue*, 56 (5).

Vigdor, Neil (2019), "Apple Card Investigated After Gender Discrimination Complaints," *The New York Times* (November 10), https://www.nytimes.com/2019/11/10/business/Apple-credit-card-investigation.html.

Vincent, James (2019), "Google and Microsoft Warn Investors That Bad AI Could Harm Their Brand," *The Verge* (February 11), https://www.theverge.com/2019/2/11/18220050/google-microsoft-ai-brand-damage-investors-10-k-filing.

Vizard, Sarah (2017), "The Brand Safety Fallout: Three in Four Marketers Say Brand Reputation Has Taken a Hit," *Marketing Week* (September 26), https://www.marketingweek.com/2017/09/26/brand-safety-fallout/.

Waytz, Adam, John Cacioppo, and Nicholas Epley (2010), "Who Sees Human? The Stability and Importance of Individual Differences in Anthropomorphism," *Perspectives on Psychological Science*, 5 (3), 219–32.

Waytz, Adam, Joy Heafner, and Nicholas Epley (2014), "The Mind in the Machine: Anthropomorphism Increases Trust in an Autonomous Vehicle," *Journal of Experimental Social Psychology*, 52, 113–17.

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